

Analysis of Favorable Injection-production Area Near the Fault Where the Directional Wells Were Deployed

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Abstract. Through the large-scale application of directional well technology, X development zone has achieved good potential tapping effect near the fault. However, with the increase of production batches of directional wells, the proportion of high-yield wells decreases, and the difficulty of well location optimization increases gradually. In order to further improve the recovery near the fault, under the condition of large-scale development of sand body in the study area, this paper focuses on the analysis of the deployment area conducive to the high yield of directional wells from the perspective of injection-production relationship. Combined with the development practice of 48 directional wells that have been deployed and have achieved good development results, the well distribution effect is evaluated from three aspects of well pattern, well spacing and injection-production location, and the deployment principle of directional wells are determined. The favorable injection-production area for the deployment of directional wells near the fault is summarized, which indicates the direction for further improving oil recovery near the fault in the old oilfield.

Keywords: Fault; directional well; injection-production relationship; well location deployment.

1. Introduction

X Development Zone is a structural reservoir and structural-lithologic reservoir. At present, it has entered a period of ultra-high water cut development. The remaining oil is highly scattered, only enriched locally, and it is difficult to tap potential. In the past, when the dense well pattern was deployed near the fault, in order to avoid the loss of the target layer, the designed well position was far away from the fault as far as possible, resulting in low well pattern density near the fault and imperfect injection and production relationship, which was the main remaining oil enrichment area. Since 2013, based on the results of well-seismic combined with fine fault interpretation, 48 large displacement directional wells have been deployed near the fault of X development zone, and good development results have been achieved, which broadens the technical path of oilfield potential tapping in the late stage of ultra-high water cut. In practice, it is found that the structure, reservoir and injection-production relationship jointly affect the development effect of large displacement directional wells near the fault. This paper analyzes the three aspects of well pattern, well spacing and injection-production location, and summarizes the laws that they have a positive impact on the development effect of directional wells, providing guidance for the popularization and application of large displacement directional wells.

2. Analysis and evaluation of directional well production effect

The evaluation standard of directional well production effect is established. Because of the good correlation between the initial cumulative production, the initial daily oil production, the initial water content and the predicted ten-year cumulative production of the directional well with large displacement, the evaluation standard of directional well production effect is established by determining the parameters.

The wells that have been put into operation in X development zone are divided into four categories : the first type well has the best production effect, the second type well is relatively good, the third type well meets the design requirements, and the fourth type well has relatively poor production effect and low production, which does not meet the design requirements (Table 1).

Table 1. Statistical table of well distribution effect classification for large displacement wells

classification	First-year cumulative production (million tons)	Initial daily oil production (tons)	Water content in the first year (%)	Prediction of ten years cumulative production (million tons)	well numbers (pcs)	outcome evaluation
first kind	≥ 0.3	≥ 15	< 80	≥ 1.5	7	very good
second kind	0.2~0.3	15~9	80~90	1~1.5	19	good
third kind	0.1~0.2	9~3	90~95	0.5~1	11	achieve design
fourth kind	< 0.1	< 3	≥ 95	< 0.5	5	worse

3. Analysis of injection-production relationship in directional wells

The distance between different well patterns and the target layer of directional wells, and the position of directional wells in the injection-production relationship of different well patterns were analyzed to evaluate the well distribution effect.

Correlation analysis was carried out by using the evaluation type of directional wells and the initial oil production with the distance of adjacent wells and the injection-production location (Table 2).

Table 2. Statistical table of distance between target layer and adjacent wells, injection-production location

Serial design number	well number	evaluation type	Oil production in early years	Basic oil wells	Basic wells	First infill well	first infill water well	Second infill oil wells	Second infill water wells	third infill oil wells	third infill water wells	Polymer flooding oil wells	Polymer flooding water wells
1	X1	2	0.8595	260	550 outside	262	199 edge	130	95 middle	222	313 outside		
2	X2	2	0.7251	335	662 outside	97	400 outside	132	345 edge	245	217 edge		
3	X3	1	0.6023	316	419 edge	75	334 outside	129	262 edge	208	362 edge		
	X4	2	0.5818	176	415 outside	75	272 edge	97	260 outside	195	239 edge		
4	X5	3	0.5719	220	480 edge	126	76 middle	81	315 outside	252	224 edge		
5	X6	2	0.8595	375	522 edge	208	130 middle	117	333 outside	222	686 outside		
...	...												
35	X7	4	0.1382				345 outside	98	316 outside			115	360 outside
36	X8	2	0.1288				344 outside	131	335 outside			141	349 outside
41	X9	3	0.0933			428	163 outside	102	200 outside			138	171 edge
42	X10	4	0.0908	658	183 outside	146	436 outside	343	180 outside			345	464 outside

(1)Effect of well spacing on well distribution : basic well
 The directional well and the basic well with relatively good effect are more than 200 meters, and the directional well is more than 250 meters away from the basic well.

(2) Effect of well spacing on well distribution : basic well
 The distance between primary wells has little effect on the overall effect of directional well layout. When the primary well is more than 60 meters, and the primary well is more than 100 meters, it can meet the design requirements of directional wells.

(3)Effect of well spacing on well distribution:secondary well

The distance between the secondary wells has little effect on the overall effect of directional well layout. The secondary wells are more than 60 meters, and the secondary wells are more than 100 meters, which can meet the design requirements of directional wells.

(4) Effect of well spacing on well distribution : third infill well

Third infill well within 100 meters from the directional well will affect the well distribution effect.

4. The position of directional wells with different distribution effect in injection-production well pattern

One kind of directional well is more than 200 meters away from the foundation well and is located outside the main flow line of injection and production of the basic well.

If the directional well is more than 170 meters away from the first, second and third wells, it will not affect the well distribution effect at any injection-production location. The second type of directional wells are more than 100 meters away from the foundation wells, and are located outside the main streamline of injection and production of the foundation wells; directional well is more than 130 meters from first infill well and more than 100 meters from second well; when the distance from polymer flooding wells is more than 240 meters, the well distribution effect will be relatively good. The three types of directional wells are close to oil wells or wells, and most of them are located on the main flow line of water injection. If the four types of directional wells are within 200 meters from the basic well or are located on the injection-production relationship main streamline of the basic well , the well distribution effect will become worse .

Table 3. Influence of injection-production position on well location deployment

Well deployment evaluation	Directional well type	Basic oil wells	Basic water wells	The first、second、third oil wells	The first、second、third water well
good effect	Category I, II	≥ 180	≥ 250 and Outside main stream	≥ 60	≥ 150
best effect	Third type	≥ 100	≥ 200 and Outside main stream	≥ 50	≥ 100
poor effect	Fourth type	≥ 100	≤ 200 or inside main stream	poor effect	

5. Conclusions and recommendations

1. Basic well pattern has the greatest impact on directional well deployment, while first, second and third wells have relatively small impact on directional well deployment.
2. It is recommended that directional well locations be deployed over 180 metres from the basic well and over 200 metres from the basic well and outside the main stream line.

References

1. Author,WANGHui-yun.:Technical Difficulties in Directional Well Logging [J]. Science and Technology Information Development and Economy, 2009.
2. Author,GENG Qiang.:Directional well trajectory design principles and control strategies [J].China Science and Technology Expo, 2017.
3. Author,WANG Xue-jian.: Shallow directional well continuous control drilling technology [J].Petroleum drilling technology, 2004.